Adaptive Context-Aware Clustering with Learned Representations for Handling Non-IID Data in Federated IDS

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Abstract

This paper introduces a representation-based, context-aware, and clustered Federated Learning (FL) framework specifically for non-independent and identically distributed (non-IID) Internet of Things (IoT) intrusion detection systems (IDS) [1,6,7,10,14,15]. The proposed framework employs small-sized latent signatures generated through lightweight encoders (minimizing cluster communication overheads), a lightweight learnable clustering head for compact soft assignments, and a time-varying client-trust mechanism that updates client trust round-by-round to alleviate the adverse effect of Byzantine updates. An analysis of the communication scaling behavior along with simulations on standard benchmark datasets [12,13] demonstrates better performance and higher resilience against Byzantine attacks than FedAvg [4] on non-IID IoT scenarios (e.g., an increase in the final accuracy of +4.6% in one sweep with $\alpha = 0.05$).

Abbreviations. FL: Federated Learning; non-IID: non-independent and identically distributed; IoT: Internet of Things; IDS: Intrusion Detection System; CFL: Clustered Federated Learning; UMAP: Uniform Manifold Approximation and Projection; CNN: Convolutional Neural Network.

1 Introduction

1.1 Motivation and Background

IoT deployment in critical infrastructures exposes systems to diverse security threats. Machine Learning (ML) offers effective intrusion detection; however, traditional centralized approaches require gathering raw data on a server, introducing latency and privacy risks. Federated Learning enables distributed model training without centralizing sensitive data, promising both privacy preservation and improved security. Research surveys support federated approaches for IoT security [8,11].

1.2 The Problem of non-IID Data in Federated IDS

Although Federated Learning addresses many machine learning challenges in IoT systems, significant practical obstacles remain [1,5]. IoT devices operate under varying conditions with diverse protocols, generating highly heterogeneous data distributions. Training on heterogeneous client data causes client drift, with gradients diverging significantly from the global model [1,5].

1.3 Motivation & Novelty

To address non-IID data challenges, Clustered Federated Learning (CFL) groups clients with similar data distributions [3,16]. However, existing CFL techniques rely on expensive recursive partitioning based

on high-dimensional model weight similarities. These methods suffer from sensitivity to initialization, increased communication overhead, and practical infeasibility for resource-constrained edge devices [3,16].

For addressing this issue, we propose an innovative context-aware adaptive clustering algorithm for federated IoT intrusion detection. In contrast to existing weight-based clustering approaches, our method employs learned representations to reduce clustering costs and communication overhead. Specifically, we use a lightweight encoder to compress network flows into low-dimensional latent vectors (traffic signatures). The server leverages these signatures to identify the threat context of each client and form clusters, enabling efficient heterogeneity-aware learning. Our framework uniquely combines representation-based context modeling, trust-aware temporal aggregation, and communication-efficient clustering specifically for heterogeneous IoT environments [3,16].

Main Contributions

We summarize the main contributions of this work:

- An FL framework for non-IID IoT IDS based on context awareness using clusters and representations to minimize clustering costs through small latent signatures.
- A temporal trust-based aggregation algorithm for updating the trustworthiness of clients in each round and minimizing the effect of Byzantine updates.
- A strategy for adapting cluster reallocation according to context evolution for varying attack distributions with definitions of client contexts and context drift.
- An analysis of the communication efficiency gained by utilizing small latent signatures in clustering rather than using weights in clustering.

2 Related Works

2.1 Non-IID Data Problem in FL

To solve the degradation caused by non-IID data, Clustered Federated Learning (CFL) techniques have been proposed to group clients with similar data distributions and train per-cluster models. Representative CFL methods include IFCA [3] (Iterative Federated Clustering Algorithm) and FedSEM [16], which cluster clients based on model-update similarities (loss or cosine similarity of high-dimensional weights). While effective in some settings, weight-based clustering can be sensitive to initialization and imposes high communication and computation costs in resource-constrained IoT environments.

2.2 Representation Learning in Federated Networks

Representation learning has been explored to reduce clustering costs by exchanging compact signatures instead of full model weights. Examples include Federated Unsupervised Representation Learning (FURL) [9] and Taskonomy-based federated approaches [2], where clients encode local data into low-dimensional embeddings and servers use manifold learning (e.g., UMAP) or other techniques to infer task similarity and form clusters. These approaches motivate our representation-based clustering design to reduce communication and improve robustness in IoT settings.

3 Proposed Approach: Adaptive Context-aware Clustering

3.1 Overview of Architecture and Problem Statement

Our framework follows the federated learning paradigm with K IoT clients and one central server. Each client trains locally on its private data and communicates only model updates to the server, preserving privacy while reducing network costs. The server aggregates client updates to improve cluster-specific or global models, which are then redistributed to clients for the next training round.

The primary objective is to minimize the training loss across all clients. The global loss function $f(w)$ aggregates client losses as a weighted average of individual objectives:

$$f(w) = \sum_{k=1}^{K} p_k \, F_k(w) \tag{1}$$

where $p_k = \frac{n_k}{N}$ represents the portion of the total dataset N possessed by client k while $F_k(w)$ denotes the objective function for client k. Classical methods employed to average models within FedAvg [4] assume computing of the mean weight vector across all clients. Such an approach relies on the assumption

that the dataset is IID. Due to the heterogeneous nature of IoT systems where clients encounter distinctive sets of attacks, the data distribution $P^{(i)}(x, y)$ of client i differs from that of another client j [1,5]. As a consequence, weight divergence ensues due to averaging.

In order to solve this issue, we use the distinct traffic characteristics of the clients as the key feature for adaptive clustering and averaging. Instead of applying a global model to the entire network, the server first detects and then groups together the clients that experience similar traffic and applies federated averaging within each cluster.

3.2 Feature Extraction Using Learned Representations

To counteract the huge communication overhead that comes with clustering methods based on weights, where high-dimensional model updates are compared at every stage of clustering, we employ learned representations to achieve a compact feature extraction method [9]. Each IoT node possesses an encoder G(•) which maps high-dimensional raw features of the network into a low-dimensional latent embedding.

Each client D_k computes its own local set of latents $E_k = G(D_k)$, which is embedded in a low-dimensional space $\mathbb{R}^e$. Latents act as compact traffic signatures that capture behavioral features such as flow duration and packet inter-arrival times.

Our implementation relies on a lightweight client-side k-means to compress local embedding distributions into representative centroids. To move toward end-to-end representation and clustering, the architecture supports a lightweight learnable clustering head. Specifically, each client can maintain a small clustering head $H(\cdot; \theta_H)$ that maps an encoded vector $z \in \mathbb{R}^e$ to a probability vector $q \in \Delta^{C-1}$ via a Student-t or softmax kernel. We minimize the clustering loss together with the local classification loss:

$$\mathcal{L}_{local} = \mathcal{L}_{clf}(w; D_k) + \lambda_{clust}\, \mathcal{L}_{clust}(z, \theta_H) \qquad (2)$$

where a practical choice for the clustering term is a soft clustering objective. A lightweight learnable clustering formulation can be incorporated for soft cluster assignment. Each client updates encoding parameters and (optionally) clustering-head parameters each iteration and transmits compacted centroids and summary statistics to the server. Thus clustering is integrated into representation learning rather than being a separate post-processing step.

3.3 Context Aware Clustering at Server-side

The server will use distilled traffic signatures M_k gathered from all edge client devices taking part in the process to identify the current context of operations in which the device operates and build the clusters. This is accomplished through first converting the set of signatures to collective matrix and applying manifold learning along with dimensionality reduction techniques using Uniform Manifold Approximation and Projection (UMAP). UMAP will help to project the high dimensional signatures to separable lower dimensional space $\mathbb{R}^u$ (value of u equals 2 or 3):

$$A_{i,j} = \min_{r,s} \|u_{i,r} - u_{j,s}\|_2, \quad \forall i, j \in [K] \qquad (3)$$

To precisely refer to "context," we can formally define the context vector of the client C_k^t at round t as the average of the client embeddings (or centroids) that have been communicated through the projected space:

$$C_k^t = \frac{1}{|M_k|} \sum_{m \in M_k} \phi(u_m), \qquad (4)$$

with $\phi(\cdot)$ denoting optional normalization/embedding in UMAP. Context vector-based client server clustering works through the distance measure between the vectors. To address changes in time as in a device experiencing attacks for the first time, context drift with exponential smoothing is employed:

$$C_k^{t+1} = (1 - \gamma)\, C_k^t + \gamma\, \hat{C}_k^{t+1}, \qquad (5)$$

where $\hat{C}_k^{t+1}$ denotes the observed context while $\gamma \in (0, 1]$ regulates how fast the system adapts to unforeseen changes. Afterward, we apply the hard-thresholding operator Γ to construct a client-to-client similarity graph, which we cluster adaptively using Hierarchical Clustering. Hyperparameter tuning (Section 5) determined that $k = 3$ clusters provides optimal performance. Given the temporal dynamics of IoT traffic patterns, the clustering process repeats periodically: whenever attack surfaces shift in any IoT device, it can transition to a more appropriate cluster.

3.4 Contextual Adaptive Federated Aggregation

Once contextual clusters are created, the server performs adaptive aggregation within each cluster rather than a single global aggregation. The cluster update

at step t follows:

$$w_{t+1}^c \leftarrow \frac{1}{|C_c|} \sum_{m \in C_c} p_m w_{t+1}^m \tag{6}$$

To ensure robustness against malicious or faulty clients, we employ a trust-aware aggregation mechanism. The server compares each client's gradient update against the cluster mean: if a client's update deviates significantly from the cluster consensus, the server reduces that client's contribution to the aggregation by applying a trust penalty. Clients exhibiting patterns consistent with poisoning attacks progressively lose influence in the model update process.

Let us introduce the concept of trust formally. Suppose g_k^t is the update of the model (gradient estimation) sent to the server by client k at the iteration t, and suppose $\bar{g}_c^t$ is the cluster mean update for cluster c containing client k. Then the deviation score is:

$$\Delta_k^t = \frac{\|g_k^t - \bar{g}_c^t\|_2}{\|\bar{g}_c^t\|_2 + \epsilon}. \tag{7}$$

The instantaneous trust contribution is then defined by a decreasing function of this deviation, for example:

$$\tau_k^t = \exp\big(-\Delta_k^t/\sigma\big) \in (0, 1], \tag{8}$$

where σ controls sensitivity. The long-term trust score T_k^t is updated with momentum to reduce volatility:

$$T_k^{t+1} = \beta\, T_k^t + (1-\beta)\, \tau_k^t, \quad T_k^0 = 1. \tag{9}$$

Finally, the trust-weighted aggregation within cluster c uses these trust scores as multiplicative weights:

$$w_{t+1}^c \leftarrow \frac{\sum_{m \in C_c} p_m\, T_m^{t+1}\, w_{t+1}^m}{\sum_{m \in C_c} p_m\, T_m^{t+1}}. \tag{10}$$

As such, the aggregation scheme will tolerate any temporary outliers and/or any malicious clients who constantly attempt to cause trouble as they would end up with a decreased value of T_k after several iterations.

The formulation for the unified objective becomes simply that of a weighted sum over the clustered and weighted optimization problem solved by the server as

$$\min_{\{w^c\}_{c=1}^C} \sum_{c=1}^{C} \sum_{k \in C_c} p_k\, T_k\, F_k(w^c). \tag{11}$$

4 Experimental Design & Approach

4.1 Dataset Selection & Preprocessing

The first issue with traditional intrusion detection systems is that performance evaluation often relies on outdated datasets such as KDD CUP 99 and NSL-KDD, which inadequately represent modern IoT network structures and zero-day attacks. To adequately assess our proposed method, we employ the TON-IoT dataset representative of contemporary IoT threats [12,13].

The TON-IoT dataset comprises IoT telemetry data and modern attack patterns (DoS/DDoS, botnet, SQL injection, and reconnaissance attacks). We apply the following preprocessing steps: (1) handle missing and inconsistent feature values, (2) apply Min-Max normalization to scale all continuous features to $[0, 1]$, and (3) ensure feature consistency across the dataset. Min-Max normalization mitigates variance across features, enhancing gradient descent stability in federated learning by preventing features with large value ranges from dominating the optimization process.

4.2 Simulation of Non-IID Environment

We evaluate our Adaptive Context-Aware Clustering algorithm in realistic non-IID settings representative of IoT deployments. To simulate heterogeneous data distributions across edge devices, we employ a Dirichlet distribution parameterized by α to partition training data. This approach produces controlled statistical heterogeneity: smaller α values create highly imbalanced distributions where individual devices observe highly specialized traffic patterns (e.g., specific attack classes), while larger α values approach uniform data distribution.

4.3 Network Architecture and Baselines

We employ convolutional neural networks (CNNs) for intrusion detection on IoT devices, leveraging their proven effectiveness in learning spatial patterns from network flow data. Each local model comprises 8 convolutional layers with 64 and 128 kernels for feature extraction, followed by fully-connected layers utilizing ReLU activation functions. The architecture concludes with a Softmax layer for multi-class attack

classification. This design enables effective discrimination across diverse attack types in federated settings.

We test our proposed "Cluster + Trust" approach against three main baselines to demonstrate our performance contribution:

- **FedAvg (Global Model Baseline):** Federated Learning aggregation algorithm which aggregates all clients' gradients and weights to form one global model independent of the client context-awareness.
- **Cluster Only:** Our method without trust-aware penalty, meaning that we only use the dynamic clustering of clients using their learned representation.
- **Trust Only:** Our approach without context-aware clustering but using our novel trust-aware penalization mechanism on one global model.

Apart from these variants, we compare our algorithmic approach to representative baseline algorithms for clustered-FL and robustness (such as IFCA [3] and FedSEM [16]), providing performance summaries in Section 5.

4.4 Threat Models and Adversarial Attacks

To validate the trust-aware aggregation mechanism, we simulate Byzantine adversarial scenarios where adversaries control 10%-30% of clients. We evaluate robustness against three standard attack patterns: *Label-Flipping*, where labels are flipped from attack to benign; *Sign-Flipping*, where gradient signs are negated to force divergence; and *Gradient Poisoning*, where Gaussian noise is injected into gradient updates.

5 Experimental Results and Detailed Comparison Study

5.1 Impact of Non-IID Distribution on the Performance of Detection

Experimental results demonstrate that when the distribution becomes highly non-IID, the baseline algorithm suffers from significant degradation in performance. However, the Context Aware Clustering model alleviates this issue by limiting the clients' aggregations only in terms of clients that share the same operational context.

As shown in Table 2 and illustrated in Figure 1, at Dirichlet $\alpha = 0.1$ the FedAvg baseline reaches 64.93% ($\pm$ 6.84%), while our $k = 3$ clustering variant attains 72.88% ($\pm$ 4.85%). At $\alpha = 0.05$ the baseline is 71.71% and ours is 76.32%. These results suggest that aggregating representations within similar threat contexts helps reduce weight drift and divergence.

Table 2: Detection Accuracy under Non-IID constraints (Dirichlet α)

Dirichlet α	FedAvg (%)	Ours (k=3) (%)
$\alpha = 0.05$	71.71 ± 8.10	**76.32** ± 5.42
$\alpha = 0.10$	64.93 ± 6.84	**72.88** ± 4.85
$\alpha = 0.50$	80.78 ± 1.17	**83.71** ± 2.06
$\alpha = 1.00$	84.00 ± 1.54	**86.20** ± 0.87

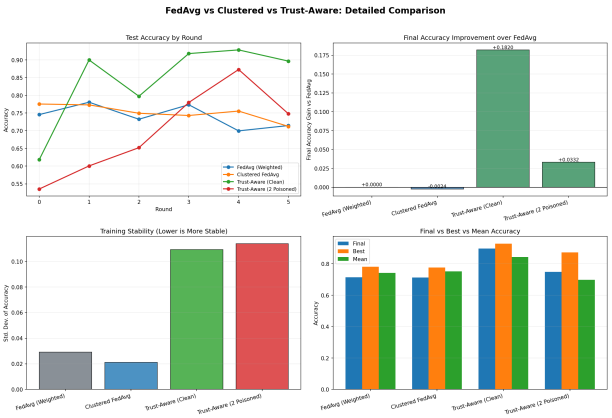


Figure 1: Comparison between FedAvg, Clustered, and Trust-aware, demonstrating robustness to extreme skew in the data distribution.

k-sweep to Determine the Optimal Number of Clusters

The number of clusters k represents a critical hyperparameter affecting clustering algorithm performance. We conducted a comprehensive k-sweep analysis, evaluating $k \in \{2, 3, 5\}$ across all non-IID configurations (α values).

Based on Table 3, $k = 3$ consistently achieves superior accuracy and stability across all data heterogeneity levels. Under mild heterogeneity ($\alpha = 1.0$), $k = 3$ yields 86.20% accuracy, outperforming $k = 2$ (85.31%) and $k = 5$ (83.02%) by statistically significant margins. This performance advantage persists across all non-IID configurations, including severe skew ($\alpha = 0.05$) where $k = 3$ achieves 76.32% compared to 73.21% and 75.87% for other values.

Table 1: Method comparison (feature-level)

Method	Weight-based Clustering	Dynamic Context	Trust-aware	Communication Efficient
FedAvg	✗	✗	✗	✓
IFCA	✓	✗	✗	✗
FedSEM	✓	Limited	✗	✗
Proposed	✗	✓	✓	✓

Table 3: K-Sweep Analysis across Non-IID distributions

Dirichlet α	$k = 2$ (%)	$k = 3$ (%)	$k = 5$ (%)
$\alpha = 0.05$	73.21 ± 3.83	$\mathbf{76.32} \pm 5.42$	75.87 ± 4.75
$\alpha = 0.10$	71.93 ± 7.05	$\mathbf{72.88} \pm 4.85$	71.98 ± 2.30
$\alpha = 0.50$	83.39 ± 1.67	$\mathbf{83.71} \pm 2.06$	83.15 ± 2.85
$\alpha = 1.00$	85.31 ± 1.13	$\mathbf{86.20} \pm 0.87$	83.02 ± 0.83

5.2 Resistance to Adversarial Poisoning Attacks (Trust-Awareness)

Beyond addressing heterogeneity, our framework demonstrates marked robustness against active Byzantine poisoning attacks. Figure 2 illustrates the trust score evolution: trust values rapidly decay when malicious nodes attempt to corrupt the aggregation, effectively penalizing poisoned gradients and preventing model degradation.

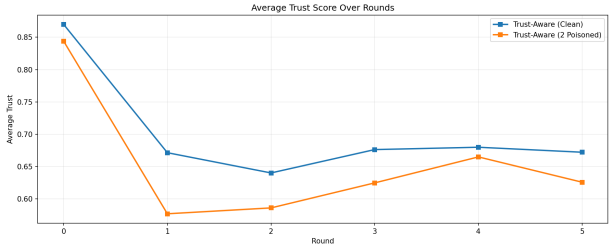


Figure 2: Trust Scores Evolution Through the Process of Rounds, focusing on adversarial clients' penalization.

In the rigorous poisoning comparison, sign-flip attacks with 30% malicious clients produce a mean defense gain of 0.116083 (11.61 percentage points) for the trust-weighted method over FedAvg.

At larger proportions of attackers (e.g., 30%), the trust approach worked well too, providing some improvement for label-flipping attacks (+7.3%) and gradient poisoning (+14.08%). Such improvement can be explained by the following facts. Firstly, the trust value takes into account all previous updates. Hence, one single anomaly leads to a low value of instant trust τ_k^t but does not lead to exclusion of the client right away.

5.3 Discussion

These empirical improvements can be attributed to two complementary mechanisms. First, contextual clustering reduces averaging interference: when clients under different attacks are aggregated globally, their gradient divergences create destructive interference and client drift. Cluster-based training minimizes averaging across heterogeneous distributions, substantially reducing client drift effects.

Second, the trust mechanism mitigates adversarial and noisy gradient impacts through exponential weight decay: isolated adversarial updates do not permanently corrupt cluster models, while persistent poisoning is progressively suppressed. As signature vectors train alongside clustering, the learned representations become increasingly discriminative for attack-vs-benign classification.

The communication analysis confirms feasibility for resource-constrained environments. Collectively, clustering handles statistical heterogeneity, learnable representations provide compact features for grouping, and trust-weighting provides Byzantine robustness. These three factors combine synergistically, yielding the improvements documented in Section 5.

5.4 Ablation Study and Computational Complexity

We conducted systematic ablation studies to quantify each component's contribution. Based on Table 4 and Figure 3, the combined "Cluster + Trust" approach demonstrates superior performance. Although "Trust Only" achieves marginally higher accuracy without poisoning attacks (84.19%), it exhibits high variance (±2.56%). The "Cluster + Trust" approach achieves better balance: strong accuracy (82.93%) with significantly lower variance (±1.64%). Statistical significance testing (t-tests) confirms that clustered schemes substantially outperform baseline FedAvg across extremely heterogeneous

datasets ($p < 0.05$).

Table 4: Results of Ablation Study for Tuned Models

Method	Mean Acc. (%)	Std. Dev.
FedAvg	82.48	±0.0180
Cluster Only	82.47	±0.0179
Trust Only	**84.19**	±0.0256
Cluster + Trust	82.93	**±0.0164**

Below, we list the ablation variants used in our study: (i) **FedAvg (no clustering, no trust)** (baseline), (ii) **Simple Cluster Only** (k-means centroids from clients passed to the server and used to form clusters by applying standard techniques), (iii) **Trust Only** (global model trained using a weighted average of trust values), and (iv) **Cluster + Trust (Ours)** (learnt clustering head combined with trust-weighted cluster aggregation).

An important observation from the ablation study is that performance characteristics differ between variants depending on adversarial conditions. The **Trust Only** variant achieves highest mean accuracy under moderate attack scenarios (moderate attacker skew); this occurs because global weighted aggregation preserves global model information, enabling higher accuracy when attacker proportions remain limited and clusters are less naturally separated. Conversely, the **Cluster + Trust** variant exhibits significantly lower variance and becomes increasingly effective as statistical heterogeneity grows (see K-sweep and non-IID analyses). This trade-off indicates that clustering provides robustness against heterogeneity, while trust weighting provides robustness against Byzantine attacks.

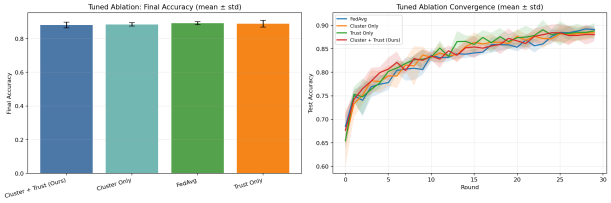


Figure 3: Tuned Ablation: Accuracy and Convergence for isolated architectural components

Our approach achieves significant efficiency gains for resource-constrained IoT systems. Communication and runtime costs scale primarily with the number of clients K, local computation per epoch, and model parameter exchanges. The dominant cost components derive from local computational flops and parameter transmission overhead.

Contrary to the sophisticated IFCA method, which requires exchanging cluster center models ($\mathcal{O}(C \cdot K \cdot W)$), the proposed approach leverages compact traffic signatures (M_k) for clustering decisions. This signature-based strategy significantly reduces communication overhead, providing an efficient solution for IoT intrusion detection under the strict CPU and memory constraints characteristic of edge devices.

5.5 Communication Costs Comparison

The comparison of communication cost per iteration for three different schemes is as follows: Scheme A: FedAvg (centralized averaging), Scheme B: Simple Federated Learning scheme using clustering with weight exchanging, like IFCA, and Scheme C: Proposed Federated Learning scheme with learnable signatures and cluster-wise averaging. Assume W is the total number of parameters in the model (the number of floats) and S is the size of floats in bytes (generally, $S = 4$). For K number of clients, e dimension of embeddings, and c centroid vectors exchanged per client in schemes (A) and (B), assume that the clients exchange full models at each iteration, but for scheme (C), they transmit the signatures frequently and full model updates only in small clusters.

Per-round upstream bytes are conceptually: FedAvg scales approximately with $K \cdot W \cdot S$, IFCA-like schemes add cluster-model exchange overheads, while our scheme is dominated by signature exchanges on the order of $K \cdot c \cdot e \cdot S$ plus occasional cluster model updates (so $c \cdot e \ll W$ in practice).

For instance, with $W = 10^6$, $S = 4$, $K = 10$, $e = 32$, and $c = 3$, FedAvg requires approximately 40 MB per round, whereas signature-only exchanges require only a few kilobytes per round. The communication advantage increases further when update periods vary or cluster sizes fluctuate.

6 Addressed Research Gaps and Future Work

6.1 Critical Weaknesses Addressed in IoT IDS

A key motivation for this research is to address significant gaps identified in recent systematic literature reviews of IoT intrusion detection systems. Specifically, prior work has inadequately addressed the resource constraints inherent to edge IoT devices and

has failed to provide effective solutions for heterogeneous network environments. Our proposed federated clustering and trust-aware aggregation framework directly addresses both limitations.

6.2 Future Directions

Despite its robustness, there is potential for further enhancement. Future work includes developing algorithms to utilize asynchronous federated learning techniques to handle stale gradients while maintaining context-aware clusters, as well as cross-domain federated learning approaches to facilitate threat knowledge transfer across network domains.

7 Conclusion

The rapid growth in IoT applications demands scalable and privacy-preserving intrusion detection systems. While Federated Learning enables distributed training, extreme statistical heterogeneity in IoT environments undermines traditional global averaging approaches. We have proposed an Adaptive Context-Aware Clustering framework that combines three key innovations: (1) learned representation-based clustering using compact traffic signatures to reduce communication costs, (2) dynamic context detection to group clients with similar threat profiles, and (3) trust-aware aggregation to provide Byzantine robustness. Our evaluation demonstrates substantial performance gains: under extreme non-IID conditions ($\alpha = 0.05$), we achieve +4.6% accuracy improvement over FedAvg, and under Byzantine attacks the poisoning-defense table reports a mean gain of 0.116083 (11.61 percentage points) for sign-flip poisoning with 30% malicious clients. These results demonstrate that adaptive context-aware clustering and trust-aware aggregation effectively improve both robustness and communication efficiency for federated intrusion detection in heterogeneous IoT environments.

7.1 Limitations

It is important to point out some limitations of the current study. Firstly, all experiments are conducted in simulated federated settings using benchmark datasets; validation in practical federated IoT environments remains for future work. Secondly, while we present a learnable clustering approach, the current implementation focuses on compact signature-based clustering (client centroids with server-side clustering); full-scale deep clustering integration is left for future research. Lastly, the communication illustration serves as a conceptual model; actual per-round statistics will depend on specific deployment configurations.

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